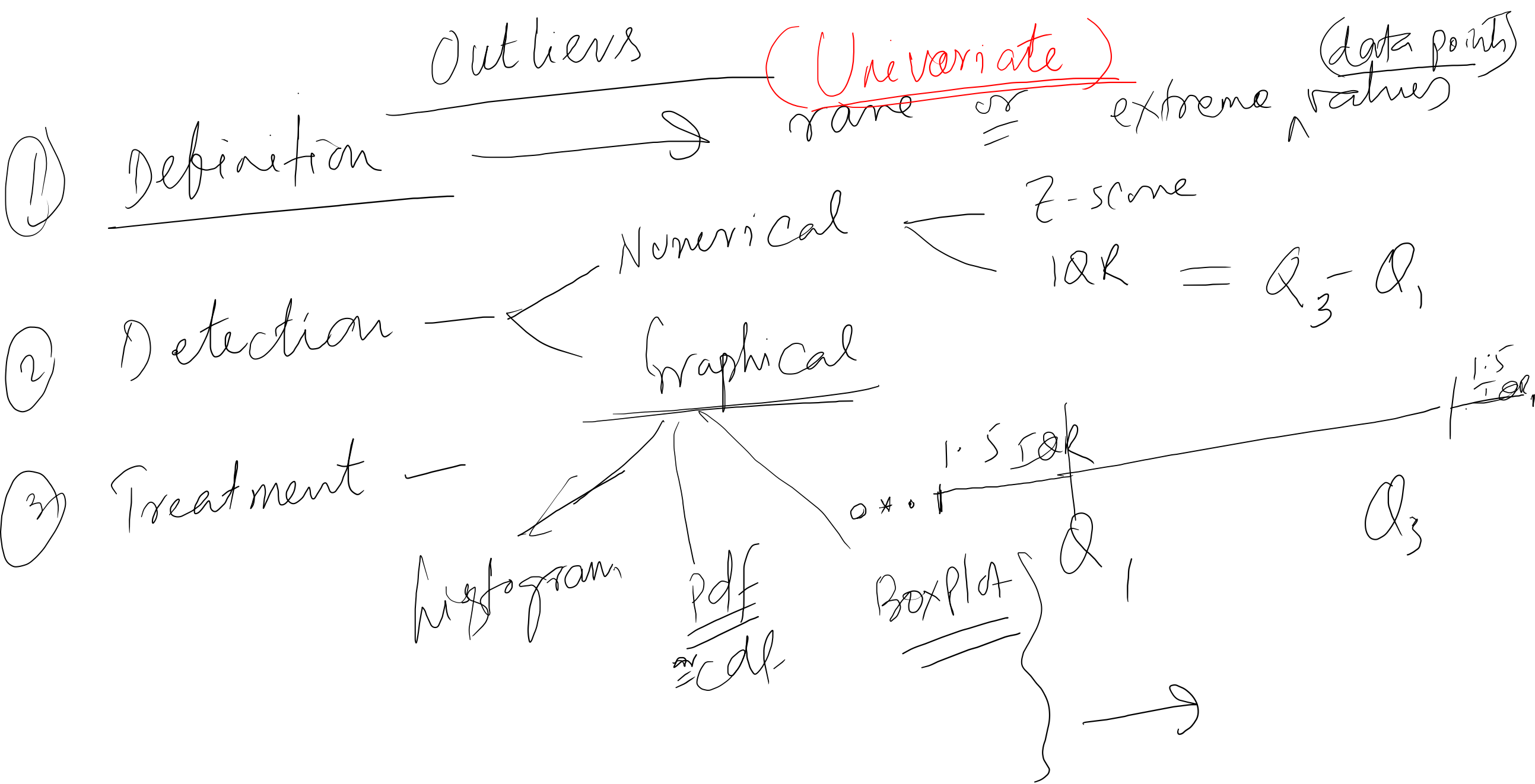




Multivariate

IQR

extend it to

Correlation matrix

Graphical

→ Heatmap

→ Scatter plots

→ Pair plots

Complex / high
dimension data

PCA

LOF
PYOD
AOD

Treatment

Remove →

Replace →

Imputation →

Range random

median

mean

percentile

Numeric data

Winsorization

Trimming

Use specific ML
algorithms

Missing values analysis

Missing Data Mechanism (MDM)

Defⁿ:-

Absence of

data values

← absence
inconsistent
diff. forms

Cases:-

may n't exist

error

Rubin (1976)

Classification:

Detection:

Treatment:

classification

MCAR - missing Completely At Random

MAR - Missing At Random

~~MAR~~ / MNAR - Not " " "

MCAR:

MAR

Y_m DEP

Y_0 X

Age	Grade	Score	Str%	Y_m
⋮				
—				—
⋮				—
—				—
⋮				—
—				—
⋮				—
—				—
⋮				—
—				—
⋮				—
—				—

Y_m IND Y_0 & other var

Missing Data Mechanism

Defn:

Cause

Types:

M C A R

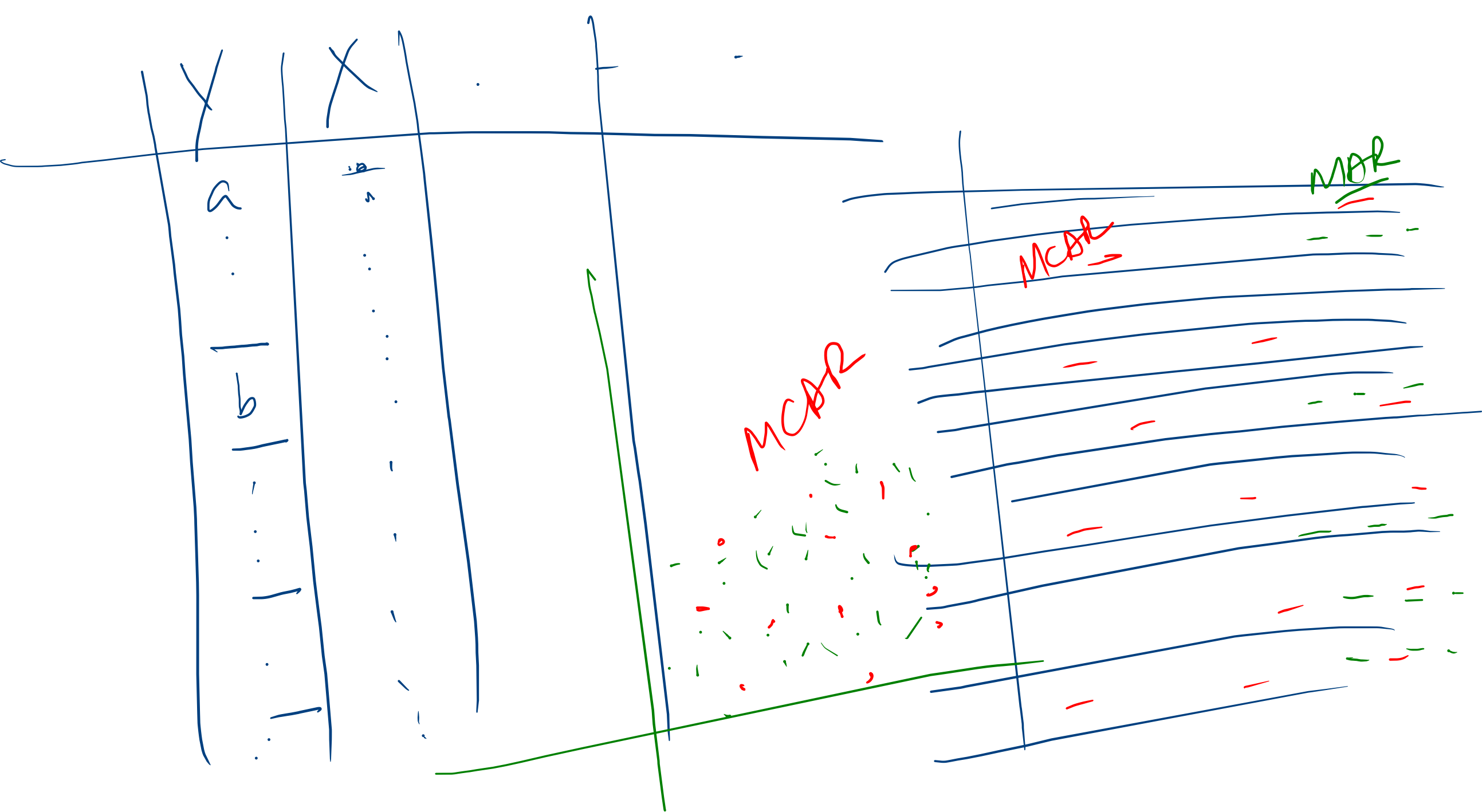
M A R

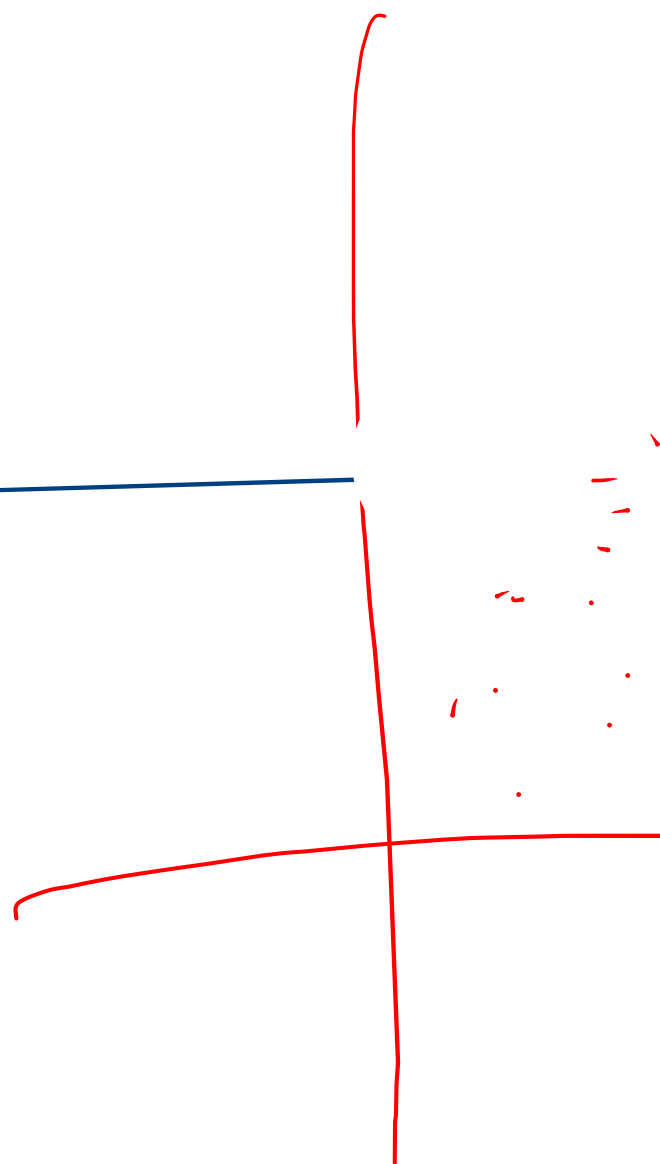
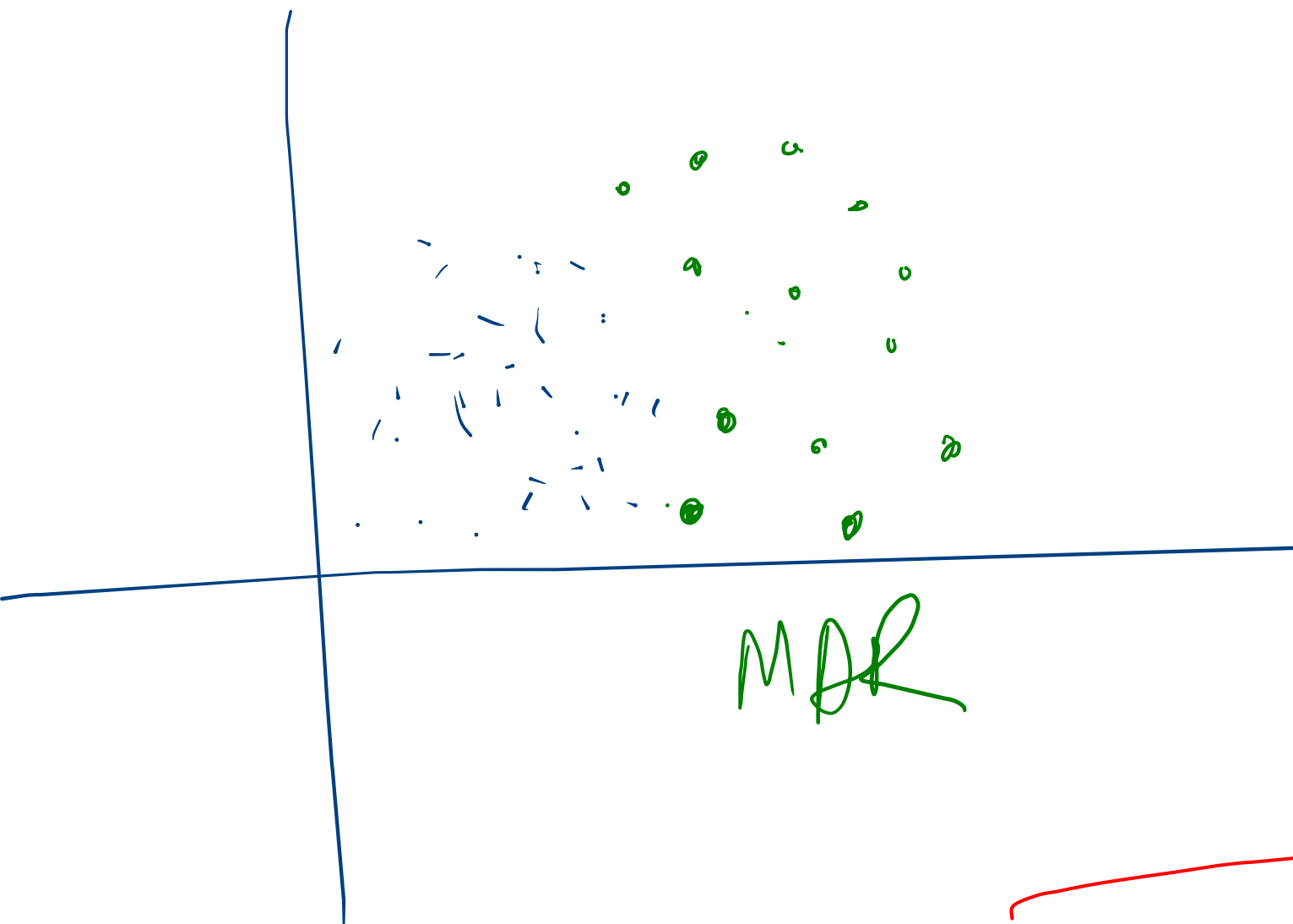
M N A R

Missing values
doesn't depend on the
observed values or missing values

Missing values
depend on
obs. values.

Missing values
depend
on
the
missing
values
itself





Detection

MCAR

Numerical



describe
is null
is na

Visual



histogram
Scatter plot

obs. values

replaced

missing no 99

Missingno Package

install

import

missingno

as msno

missingno

dendrogram

heatmap

matrix

bar

Y	R	Y _{mis}	Y _{obs}
0	1		
0	1		
.	1		
0	0		
0	1		
0	1		
0	1		

$$R = \begin{cases} 0 & \text{if value missing} \\ 1 & \text{if value observed} \end{cases}$$

$$Y = \begin{pmatrix} Y_{obs} & Y_{mis} \end{pmatrix}$$

: parameters

~~MIM in MCAR~~

$$P(R \mid Y_{\text{obs}}, Y_{\text{mis}}, \psi)$$

$$\underline{\underline{R=0}}$$

 Missing

$$P(R=0 \mid Y_{\text{obs}}, Y_{\text{mis}}, \psi) = P(R=0)$$

MAR

$$P(R=0 \mid Y_{\text{obs}}, Y_{\text{mis}}, \psi) = P(R=0 \mid Y_{\text{obs}}, \underline{\psi})$$

MNAR

$$P(R=0 \mid Y_{\text{obs}}, Y_{\text{mis}}, \psi) =$$

	V_1	V_2	V_3
V_1	1	0.1	0.01
V_2	0.1	1	0.02
V_3	0.01	0.02	1

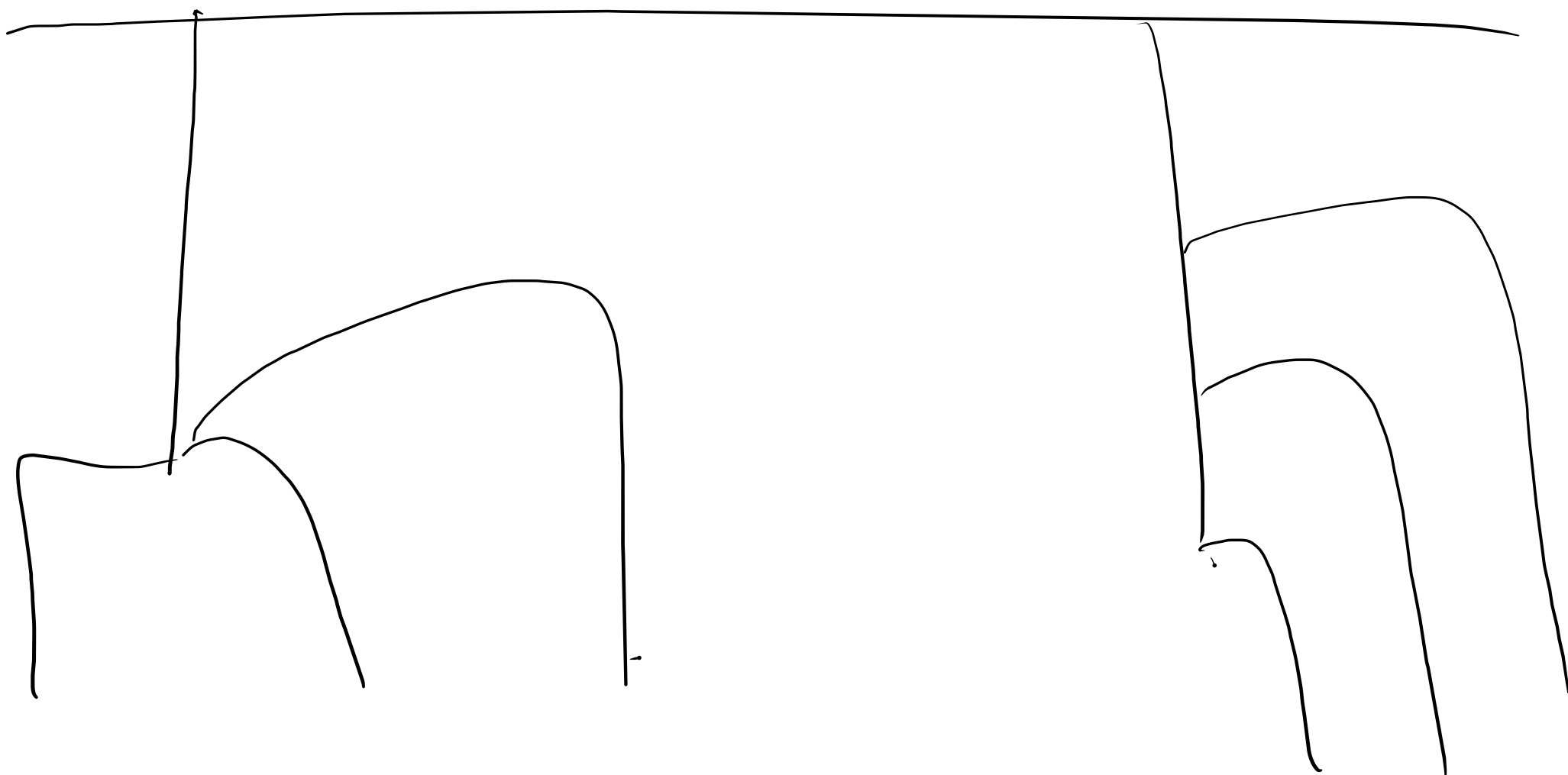
MCAR

~~MAR~~

0.4
0.6
0.3
0.1
0.2

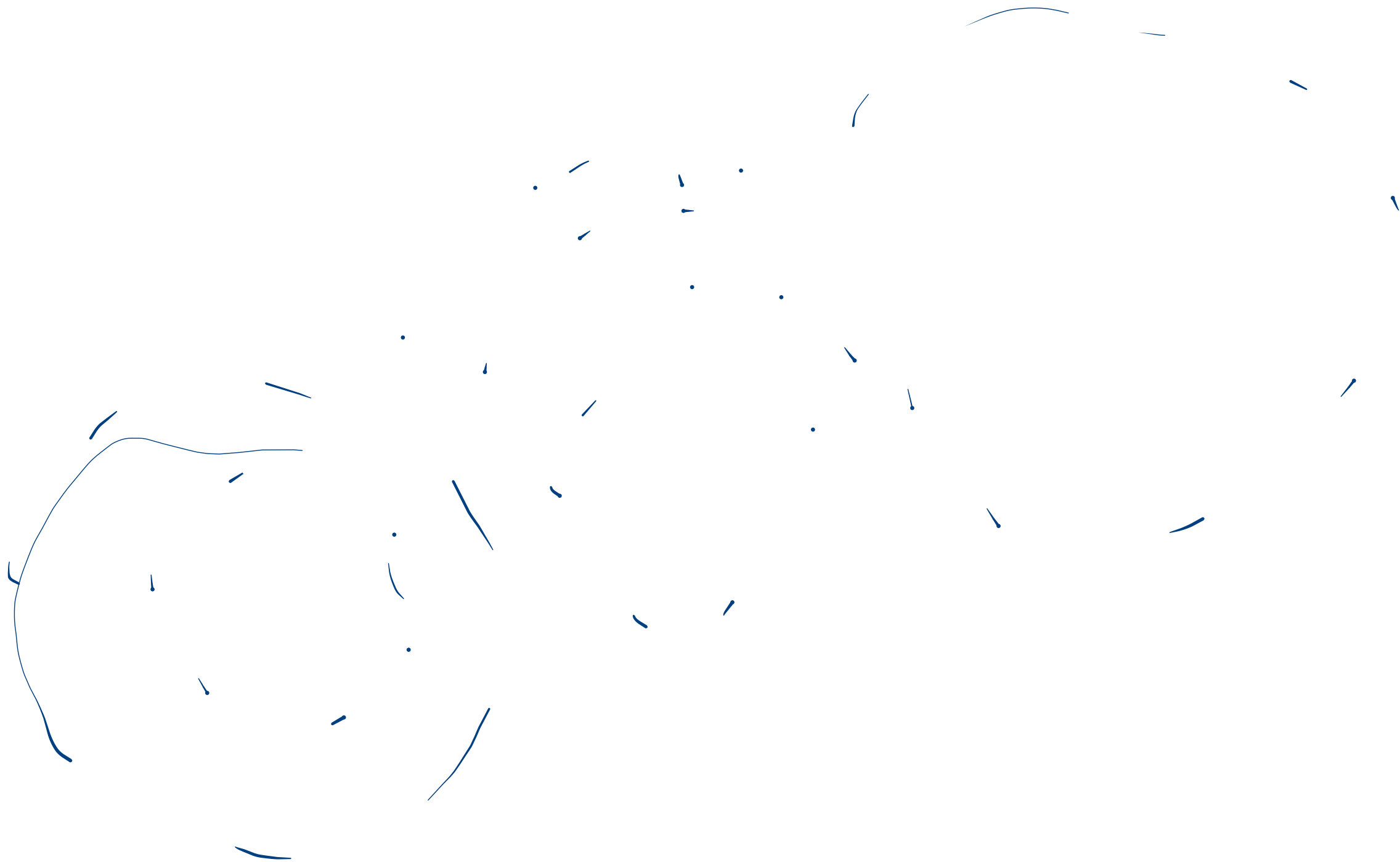
~~MAR~~

	V_1	V_2	V_3
V_1	1	0.7	0.5
V_2	0.7	1	0.5
V_3	0.5	0.5	1



WARR





MSAO

ben

mation

hydrogram

head map

non-Confirmatory Signal

Confirmatory
tests

Statistical
method / Test

①

~~Little's~~ MCR Test

(Overall test)

②

Dixon's MCR Test

(Univariate)

③

Hawkins MCR Test

(Multivariate)

Hypothesis Tests

①

Statement of the Test Hypotheses

H_0 : Data MCAR
 H_1 : Not MCAR

H_0 Null

H_1 Alternate

α

Significance Level

type of test

one-tailed
two-tailed

Test Statistic

test-statistic

Compute the observed value

χ^2

from

Sample data

p

Missing value Treatment

Missingno. py



bar
matrix

heat map

dendrogram

Confirmatory tests

MCAR

Likelihood
MCMC
test

Dixon's

Hawkins's
test

global

$H_0: \text{MCAR}$
 $H_1: \text{NOT MCAR}$

α , two-tailed test

FDA
 Univariate
 Histogram
 Q-Q plots

p

Test statistic

$\alpha = 0.05$
 S.I. $\chi^2_0 = y$
 $\chi^2_\alpha = x$

fail to reject

Reject
 $\alpha = 0.05$

$$p = 0.26$$
$$\alpha = 0.05$$

$$p > \alpha$$

→ Fail to Reject
Missing Data MCAR | H_0

$$p = 0.01$$
$$\alpha = 0.05$$

$$p < \alpha$$

→ Reject H_0
Not MCAR

MCAR

Treatment

% missing values

df. is small

* is na

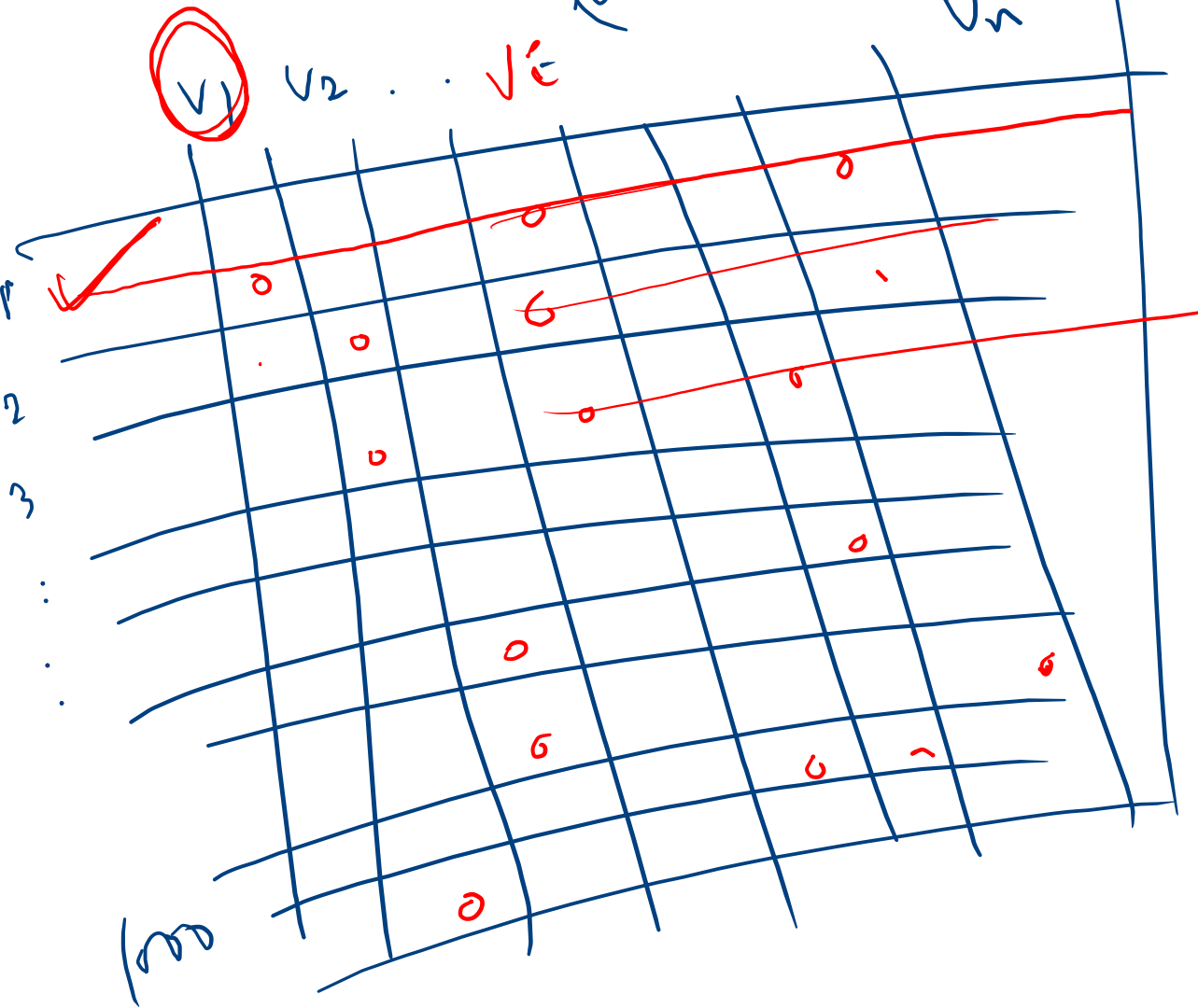
info

5-10%

Complete Case Analysis

Complete case
(list wise deletion)

Available case
(pairwise deletion)



Imputation

Mean

Median

function

Numeric
data

Regression

$$V_i = C_0 + C_1 V_2 + C_3 V_3 + E$$

Data transform

What is?

raw data

→ desired form

Why?

reducing dimension
removing outliers
comparing variables on a scale
improve model performance
interpretability ✓

When?

before modeling
use any algorithm that needs assumption

How?

Types

①

Scaling

Z-score

min-max

Robust

$X - x_{\text{median}}$
IQR

$$\begin{aligned} E\left(\frac{X - \mu}{\sigma}\right) &= \frac{1}{\sigma} E(X - \mu) \\ &= \frac{1}{\sigma} [E(X) - \mu] \\ &= \frac{1}{\sigma} (\mu - \mu) = 0 \end{aligned}$$

Z Φ

$$\frac{X - \mu}{\sigma}$$

$$\begin{aligned} \mu &= \frac{\sum x_i}{n} \\ \sigma &= \sqrt{\frac{\sum (x_i - \mu)^2}{n}} \\ s &= \sqrt{\frac{\sum (x_i - \mu)^2}{n-1}} \end{aligned}$$

$$\frac{x_{\text{max}} - x_{\text{min}}}{x_{\text{max}} + x_{\text{min}}}$$

Numeric

2

function

Power

Quantile

Normal
rank($\psi(n)$)

Uniform
 $y = \text{rank}(n)$

given data

log -

Square -

Inverse -

Square root -

Box-Cox

Yeo-Johnson

highly ^{right} Skewed, Positive values

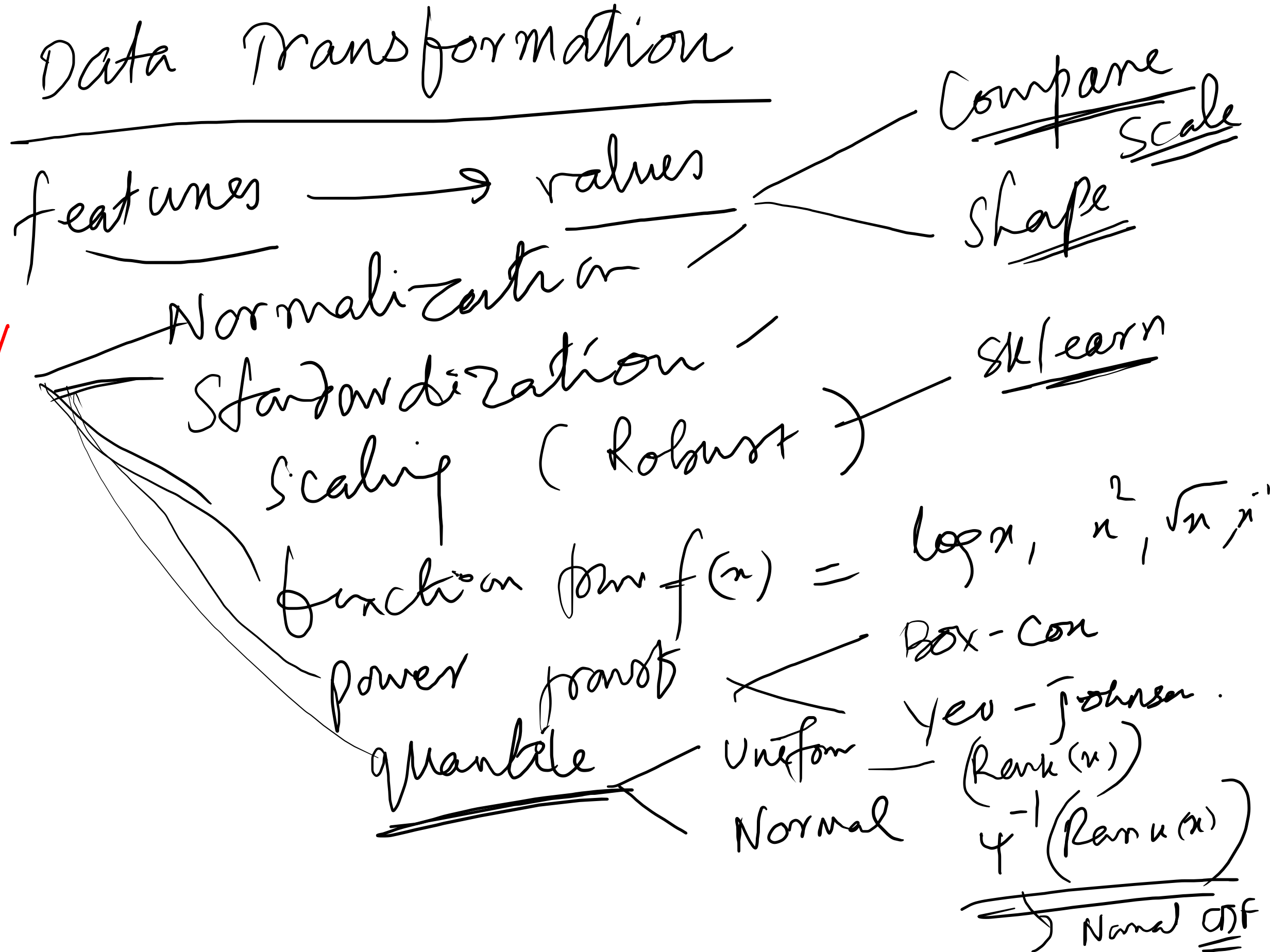
Right Skewed
Positive, Right skewed

$$y(\lambda) = \begin{cases} \frac{y^\lambda - 1}{\lambda} & \lambda \neq 0 \\ \log(y) & \lambda = 0 \end{cases}$$

$$\psi(\lambda, y) = \begin{cases} \frac{(y+1)^\lambda - 1}{\lambda} & \lambda \neq 0, y \geq 0 \\ \log(1+y) & \lambda = 0, y \geq 0 \\ -\log(1-y) & \lambda = 2, y < 1 \\ \frac{-(1-y)^{2-\lambda} - 1}{2-\lambda} & \lambda \neq 2, y < 1 \end{cases}$$

Stage 1 →

Numeric
features



Stage II

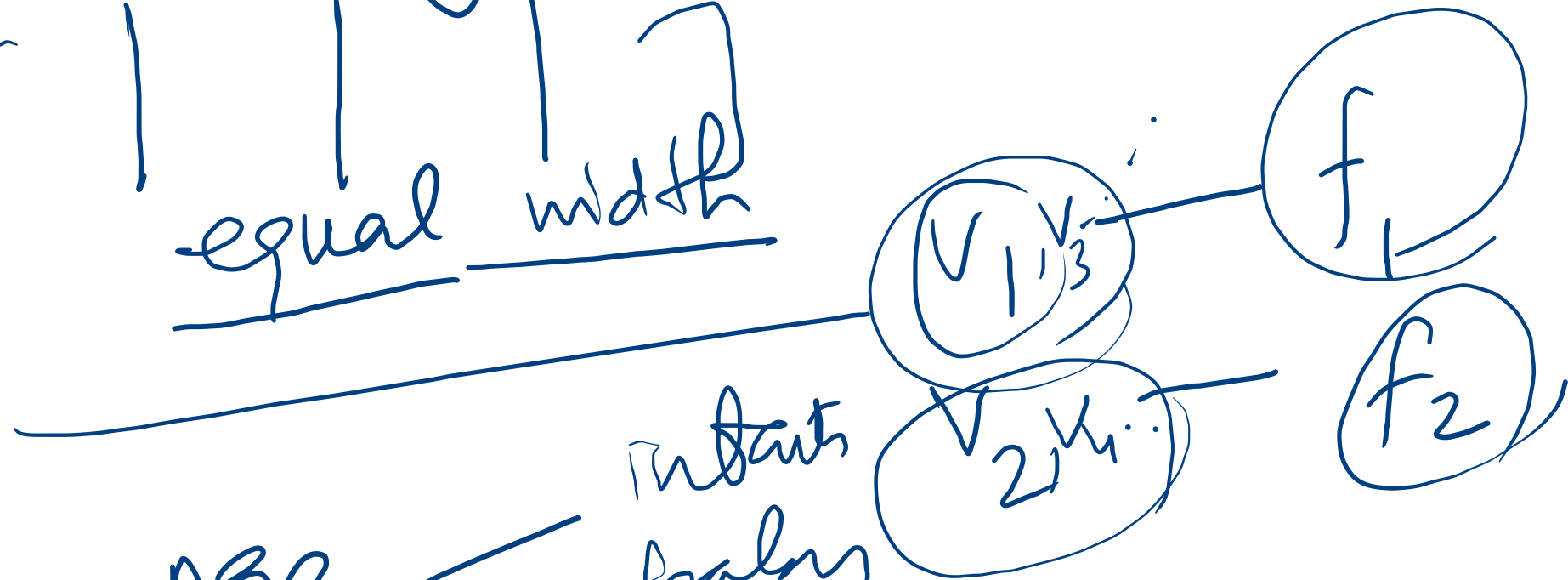
Numeric $\longrightarrow$ Category

$\longrightarrow$ Discretization / Binning

① Equal-width

[| | | |]
equal width

② Equal-frequency



③ Domain-specific

Age $\longrightarrow$ Infant
Salary

④ cluster-based

Group 4, Group 2, Group 3

Stage III

Category $\rightarrow$ Numeric

Gender

M F T

(Encoding)

one-hot

Categories

1	0	0
0	1	0
0	0	1

Label

0
1
2
3

ordinal

ordinal variable

Binary
Target

mean

0
1
2
3

00
01
10
11

Hash
entity